Synthetic-to-Real Image Translation for Chessboard Rendering

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January 15, 2026

Abstract

In this project, we address the challenge of Synthetic-to-Real image translation for chessboard rendering. Our goal is to generate realistic chessboard images from synthetic layouts while preserving geometric consistency. We propose a robust pipeline utilizing the **Brownian Bridge Diffusion Model (BBDM)** integrated with a fine-tuned **VQGAN** for latent space representation. To ensure high-quality training data, we developed a preprocessing pipeline using **ffmpeg** for frame deduplication and the **Segment Anything Model (SAM)** for removing occluded samples (e.g., hands). While initial experiments with GAN-based methods (e.g., CUT) yielded suboptimal results, our diffusion-based approach demonstrates superior visual fidelity. We present quantitative and qualitative results, including training progression across epochs, demonstrating the effectiveness of our method.

1 Introduction

1.1 Task Description and Motivation

The task involves translating synthetic chessboard images—generated via rendering engines—into photorealistic images that mimic real-world capture conditions. This is crucial for bridging the domain gap in computer vision tasks where real labeled data is scarce but synthetic data is abundant.

1.2 Challenges and Goals

Key challenges include:

- **Geometric Preservation:** The translation must not alter the position or identity of chess pieces.
- **Data Quality:** Real-world data often contains artifacts (motion blur, occlusions by hands) that can degrade model performance.

- **Limited Paired Data:** We faced a significant scarcity of high-quality paired data. The initial dataset contained approximately 500 synthetic-real pairs. After filtering out about 50 pairs containing occlusions (e.g., hands), we were left with a small training set of ∼450 clean pairs, making robust generalization challenging.

1.3 Main Contributions

Our main contributions are:

1. **Data Curation and Enhancement:** We executed a rigorous data improvement process involving three key steps:
   - We generated approximately ∼400 additional synthetic-real pairs by using **ffmpeg** to filter and extract stable frames from video data.
   - We employed **SAM** [1] to detect and remove real images containing hand occlusions, ensuring a clean training set.
   - We optimized the synthetic data generation in **Blender** by experimenting with various parameters (camera angles, lighting, and color grading) to better match the real domain distribution.
2. **Comprehensive Model Fine-Tuning:** We utilized the **BBDM** [2] architecture and performed fine-tuning on both critical components of the pipeline: the dimensionality reduction module (VQGAN latent encoder) and the Brownian Bridge diffusion module itself. ***(Note: Roni needs to review specific details here regarding the fine-tuning process).***
3. **Loss Function Optimization:** We improved the standard loss function strategy to enhance structural preservation and texture fidelity. ***(Note: Roni needs to review specific details here).***

2 Related Work

2.1 Generative Adversarial Networks (GANs)

GANs have been the standard for Image-to-Image (I2I) translation. Notable models include **Pix2Pix** for paired data and **CycleGAN** [3] or **CUT** (Contrastive Unpaired Translation) [4] for unpaired data. Below is a brief overview of these architectures:

- **Pix2Pix:** Employs a conditional adversarial network to learn the mapping from input to output images, requiring perfectly aligned image pairs during training.
- **CycleGAN:** Overcomes the need for paired data by enforcing transitivity through a cycle-consistency loss, training two mapping functions simultaneously to translate between domains.
- **CUT:** Improves upon CycleGAN for unpaired settings by using contrastive learning to maximize the mutual information between corresponding patches of the input and output images, often facilitating faster training.

We initially explored these methods but found them limited in handling the complex texture mapping required for this task.

2.2 Vector Quantized Models (VQGAN)

VQGAN [5] combines convolutional approaches with transformers by learning a discrete codebook of image constituents. It serves as a powerful perceptual compression stage, allowing subsequent models to operate in a lower-dimensional latent space without losing high-frequency details.

2.3 Diffusion Models and BBDM

Diffusion models, such as DDPM [6], have recently surpassed GANs in generation quality. Specifically, **BBDM** [2] models the translation between two domains as a stochastic process governed by a Brownian Bridge. This allows for direct image-to-image translation without requiring cycle consistency, making it highly suitable for paired synthetic-to-real tasks.

2.4 Segmentation Foundation Models (SAM)

The **Segment Anything Model (SAM)** [1] has revolutionized semantic segmentation. We leverage SAM not for generation, but as a robust filtering tool to detect and remove occlusions (such as players' hands) from our training dataset.

3 Method

3.1 Data Generation and Preprocessing

A critical part of our approach was ensuring high-quality input data. We employed a three-stage pipeline:

3.1.1 Temporal Filtering with ffmpeg

The raw data consisted of video recordings (PGN-labeled games). Since chess positions remain static for multiple seconds, extracting every frame results in high redundancy. We used **ffmpeg** to detect identical consecutive frames and extract only unique stable frames, significantly reducing dataset size while maximizing diversity.

3.1.2 Occlusion Removal with SAM

Real-world chess videos often contain hand movements during piece manipulation. Training on occluded images introduces artifacts. We utilized **SAM** [1] to detect "person" or "hand" masks. Images containing significant occlusions were automatically discarded from the training set to ensure the model learns only clean board representations.

3.1.3 Enhanced Synthetic Data (Blender)

We utilized the provided Blender script but introduced several improvements to bridge the domain gap:

- Randomized camera angles and focal lengths to match real-world variance.
- Improved lighting conditions and shadows.

- Color grading adjustments to better align with the specific color temperature of the real dataset.

3.2 Model Architecture: BBDM with VQGAN

We selected the **Brownian Bridge Diffusion Model (BBDM)** [2] as our core architecture.

- **Latent Space (VQGAN):** Instead of operating in pixel space, we use a VQ-GAN [5] to compress images into a discrete latent representation. This reduces computational complexity and focuses the diffusion process on semantic structure.
- **Fine-Tuning Strategy:** We performed fine-tuning on two levels:
  1. **VQGAN Fine-tuning:** Adapted the codebook to specifically capture chessboard textures and piece geometries.
  2. **BBDM Fine-tuning:** Trained the diffusion bridge specifically on the mapping between our synthetic renders and the cleaned real dataset.

3.3 Loss Functions

To ensure both perceptual quality and structural fidelity, we optimized a composite loss function.

$$\mathcal{L}_{total} = \lambda_{diff}\mathcal{L}_{diffusion} + \lambda_{perc}\mathcal{L}_{LPIPS} + \lambda_{struct}\mathcal{L}_{structure} \tag{1}$$

We incorporated the LPIPS metric [7] to improve perceptual similarity.

4 Experiments

4.1 Experimental Setup

- **Dataset:** [Number] pairs of Synthetic-Real images after filtering.
- **Hardware:** Trained on [GPU Name, e.g., NVIDIA A100/T4].
- **Hyperparameters:** Batch size: [X], Learning rate: [Y], Epochs: [Z].

4.2 Qualitative Results

We present the progression of our model throughout the training process. Figure 1 demonstrates how the model gradually learns to synthesize realistic textures.

4.3 Quantitative Evaluation

TODO: Add your FID/LPIPS scores here.

Placeholder for Training Progression Image

Insert an image showing: Synthetic — Epoch 10 — Epoch 50 — Epoch 100 — Final — Real

Figure 1: Training progression: Visualizing the domain translation at different epochs.

5 What Did Not Work

Prior to adopting BBDM, we experimented extensively with GAN-based approaches, specifically **CUT** [4].

- **Issue:** The GAN models struggled to maintain the precise geometry of chess pieces, often hallucinating pieces or distorting the board grid.
- **Result:** As seen in Figure 2, the outputs suffered from mode collapse and unrealistic artifacts.
- **Conclusion:** This led us to switch to Diffusion Models, which offer more stable training and better structural preservation.

Placeholder for Failed GAN Result

Insert an example of a failed CUT/CycleGAN output.

Figure 2: Example of failed translation using CUT/GAN.

6 Ablation Study

To validate our pipeline components, we analyzed the impact of:

1. **w/o SAM Filtering:** Training on raw data (including hands) resulted in the model generating ghostly artifacts over the board.

2. **w/o VQGAN Fine-tuning:** Using a generic pre-trained VQGAN resulted in blurry piece edges.

Table 1: Ablation Study Results

Configuration	**Visual Observation**
Full Pipeline (Ours)	Sharp, Realistic, Clean
w/o SAM Filtering	Artifacts, Occlusions
w/o VQGAN FT	Blurry Textures

7 Discussion and Limitations

While our BBDM approach yields high-quality results, the inference time is slower compared to single-pass GANs due to the iterative diffusion process. Future work could explore distillation techniques to speed up sampling.

8 Conclusion

We presented a robust Synthetic-to-Real translation pipeline for chessboard images. By combining careful data curation (SAM, ffmpeg) with advanced diffusion models (BBDM+VQGAN), we achieved significant improvements over traditional GAN baselines.

References

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